
Edge detection

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Machine Vision Technology							
Semantic information				Metric 3D information			
Pixels	Segments	Images	Videos	Camera		Multi-view Geometry	
Convolutions Edges & Fitting Local features Texture	Segmentation Clustering	Recognition Detection	Motion Tracking	Camera Model	Camera Calibration	Epipolar Geometry	SFM
10	4	4	2	2	2	2	2

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Edge detection

- **Goal:** Identify sudden changes (discontinuities) in an image
 - Intuitively, most semantic and shape information from the image can be encoded in the edges
 - More compact than pixels
- **Ideal:** artist's line drawing (but artist is also using object-level knowledge)

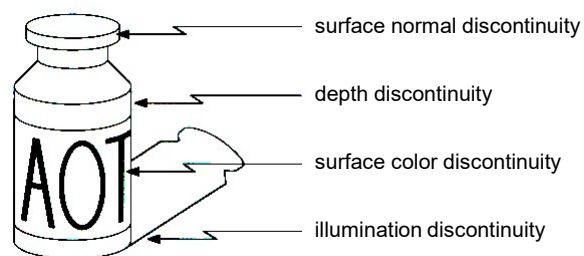


Source: D. Lowe

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Origin of edges

Edges are caused by a variety of factors:

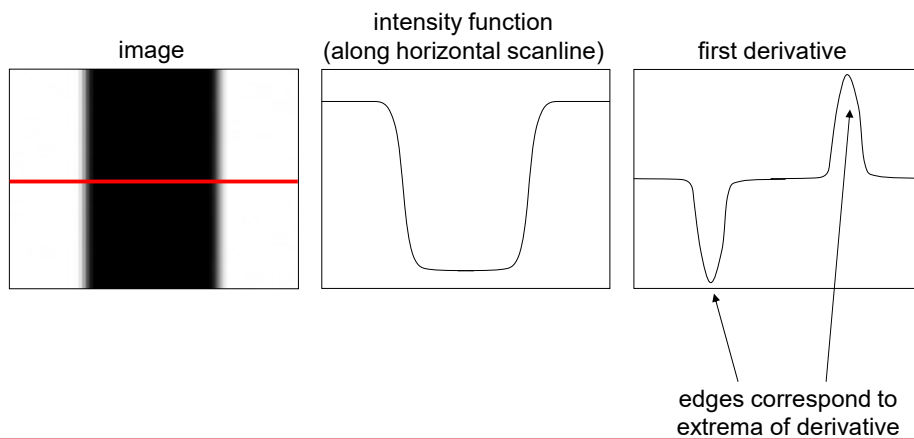


Source: Steve Seitz

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Characterizing edges

- An edge is a place of rapid change in the image intensity function



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Derivatives with convolution

For 2D function $f(x,y)$, the partial derivative is:

$$\frac{\partial f(x,y)}{\partial x} = \lim_{\varepsilon \rightarrow 0} \frac{f(x + \varepsilon, y) - f(x, y)}{\varepsilon}$$

For discrete data, we can approximate using finite differences:

$$\frac{\partial f(x,y)}{\partial x} \approx \frac{f(x+1, y) - f(x, y)}{1}$$

To implement above as convolution, what would be the associated filter?

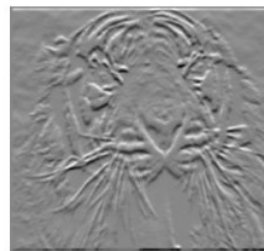
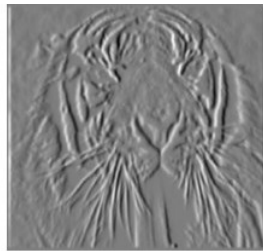
Source: K. Grauman

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Partial derivatives of an image

$$\frac{\partial f(x, y)}{\partial x}$$

$$\begin{bmatrix} -1 & 1 \end{bmatrix}$$



$$\frac{\partial f(x, y)}{\partial y}$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$



Which shows changes with respect to x?

Source: S. Lazebnik

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Finite difference filters

Other approximations of derivative filters exist:

Prewitt: $M_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$; $M_y = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$

Sobel: $M_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$; $M_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$

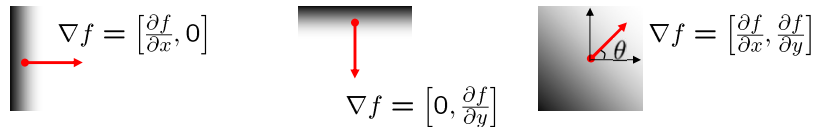
Roberts: $M_x = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$; $M_y = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Source: K. Grauman

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Image gradient

The gradient of an image: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$



The gradient points in the direction of most rapid increase in intensity

- How does this direction relate to the direction of the edge?

The gradient direction is given by $\theta = \tan^{-1} \left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x} \right)$

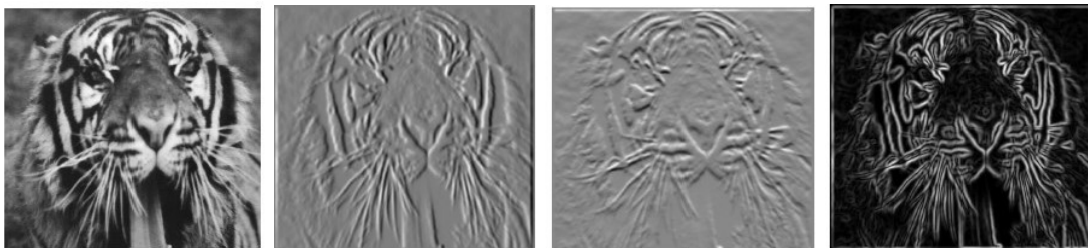
The *edge strength* is given by the gradient magnitude

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

Source: S. Seitz

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Gradient Magnitude



X-Derivative

Y-Derivative

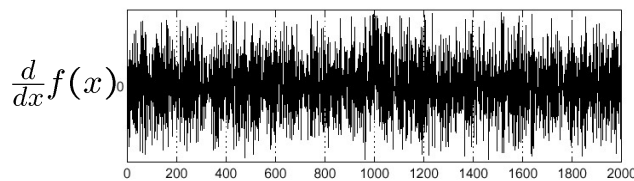
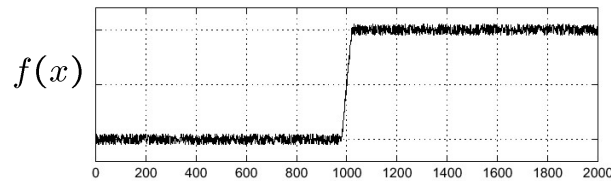
Gradient Magnitude

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Effects of noise

Consider a single row or column of the image

- Plotting intensity as a function of position gives a signal

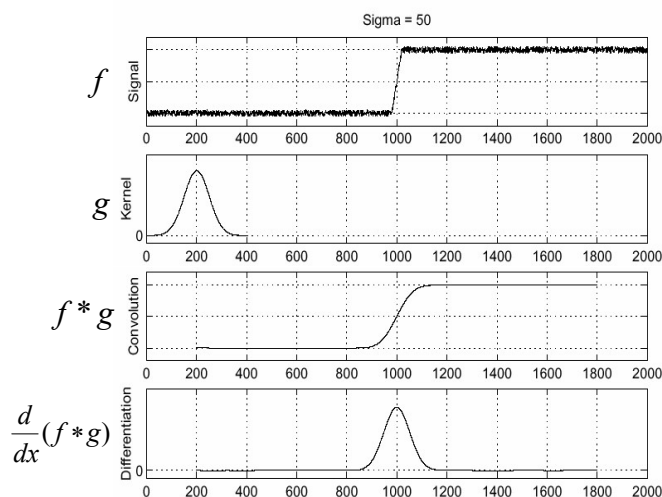


Where is the edge?

Source: S. Seitz

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Solution: smooth first



- To find edges,

look for peaks

in $\frac{d}{dx}(f * g)$

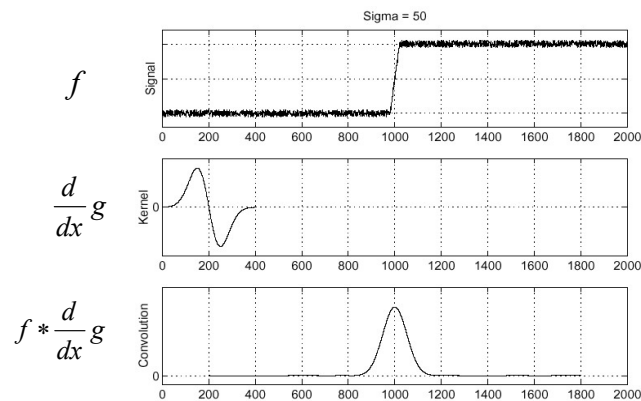
Source: S. Seitz

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Derivative theorem of convolution

- Differentiation is convolution, and convolution is associative:

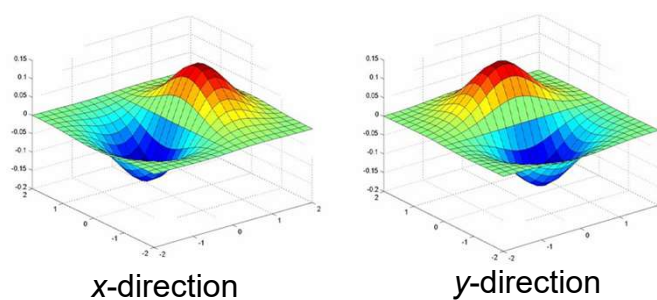
- This saves us one operation: $\frac{d}{dx}(f * g) = f * \frac{d}{dx}g$



Source: S. Seitz

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Derivative of Gaussian filter

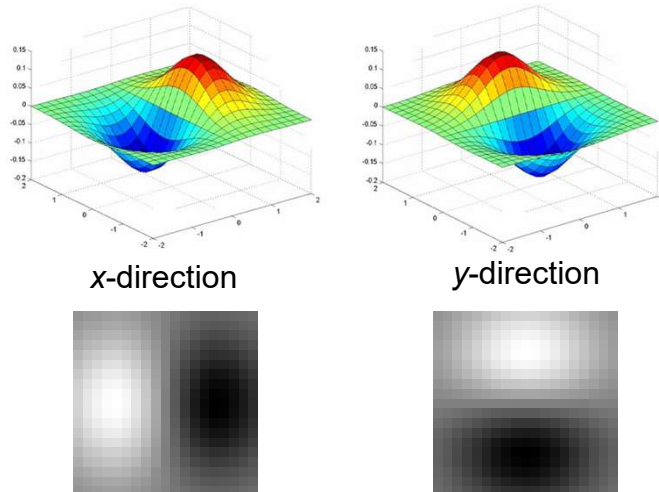


Are these filters separable?

Source: S. Lazebnik

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Derivative of Gaussian filter

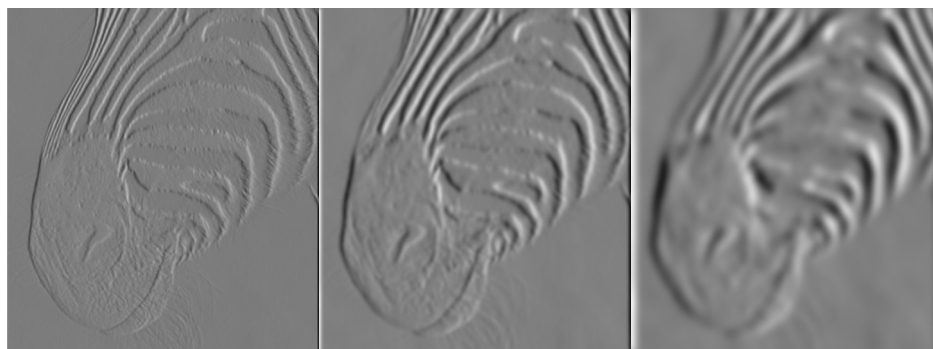


Which one finds horizontal/vertical edges?

Source: S. Lazebnik

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Scale of Gaussian derivative filter



1 pixel

3 pixels

7 pixels

Smoothed derivative removes noise, but blurs edge.
Also finds edges at different “scales”

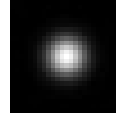
Source: D. Forsyth

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Review: Smoothing vs. derivative filters

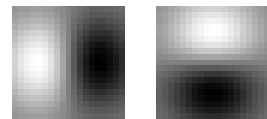
Smoothing filters

- Gaussian: remove “high-frequency” components; “low-pass” filter
- Can the values of a smoothing filter be negative?
- What should the values sum to?
 - **One**: constant regions are not affected by the filter



Derivative filters

- Derivatives of Gaussian
- Can the values of a derivative filter be negative?
- What should the values sum to?
 - **Zero**: no response in constant regions
- High absolute value at points of high contrast



Source: S. Lazebnik

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The Canny edge detector



original image

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The Canny edge detector

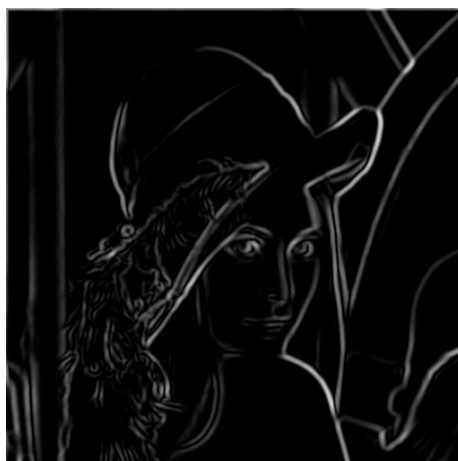


norm of the gradient

Source: S. Lazebnik

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The Canny edge detector

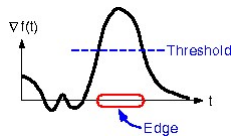
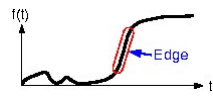


thresholding

Source: S. Lazebnik

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The Canny edge detector



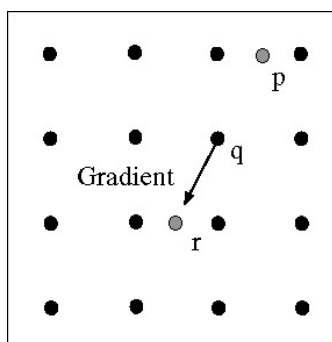
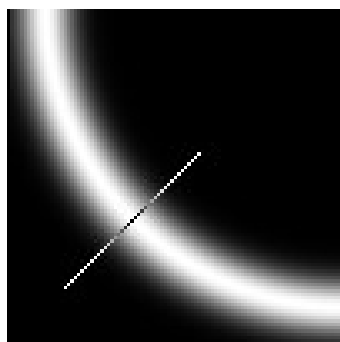
How to turn these thick regions of the gradient into curves?

thresholding

Source: S. Lazebnik

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Non-maximum suppression



Check if pixel is local maximum along gradient direction, select single max across width of the edge

Source: S. Lazebnik

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The Canny edge detector



Problem:
pixels along
this edge
didn't
survive the
thresholding

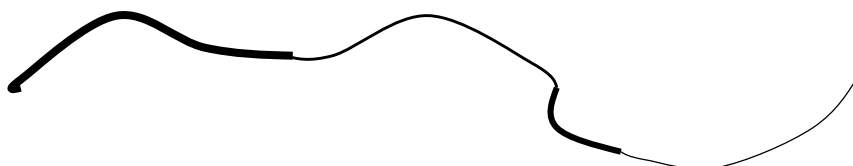
thinning
(non-maximum suppression)

Source: S. Lazebnik

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Hysteresis thresholding

Use a high threshold to start edge curves, and a low threshold to continue them.



Source: S. Seitz

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Hysteresis thresholding



original image



high threshold
(strong edges)



low threshold
(weak edges)



hysteresis threshold

Source: L. Fei-Fei

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Recap: Canny edge detector

1. Filter image with derivative of Gaussian
2. Find magnitude and orientation of gradient
3. **Non-maximum suppression:**
 - Thin wide “ridges” down to single pixel width
4. **Linking and thresholding (hysteresis):**
 - Define two thresholds: low and high
 - Use the high threshold to start edge curves and the low threshold to continue them

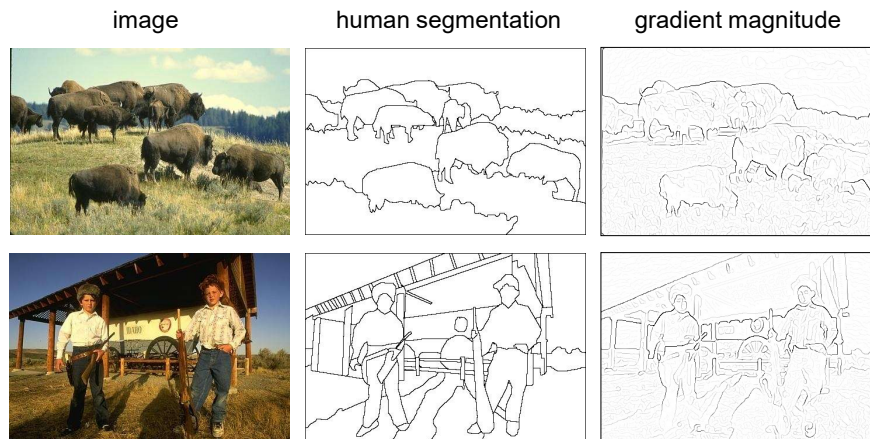
MATLAB: `edge(image, 'canny');`

J. Canny, [*A Computational Approach To Edge Detection*](#), IEEE Trans. Pattern Analysis and Machine Intelligence, 8:679-714, 1986.

Source: S. Lazebnik

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Edge detection is just the beginning...



Berkeley segmentation database:

<http://www.eecs.berkeley.edu/Research/Projects/CS/vision/grouping/segbench/>

Source: S. Lazebnik